A

Project Report

SMART WATER CONSUMPTION MONITORING DASHBOARD

on

Submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

By

V.MIDHUNA

24K81A05R2

Under the guidance of

Ms.V.SIRI

Assistant professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

St. MARTIN'S ENGINEERING COLLEGE

SHAPING VIKSIT BHARAT'S BRIGHTEST MINDS

UGC Autonomous

Affiliated to JNTUH, Approved by AICTE

Accredited by NBA & NAAC A+,ISO 9001-2008 Certified

Dhulapally,Secunderabad-500 100

www.smec.ac.in

St. MARTIN'S ENGINEERING COLLEGE

SHAPING VIKSIT BHARAT'S BRIGHTEST MINDS

UGC Autonomous

NBA& NAAC A+ Accredited Dhulapally, Secunderabad - 500 100

www.smec.ac.in

Certificate

This is to certify that the project entitled "**SMART WATER CONSUMPTION MONITORING DASHBOARD** " is being submitted by **V.MIDHUNA (24K81A05R2)** in fulfilment of the requirement for the award of degree of **BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING** is recorded of Bonafide work carried out by them. The result embodied in this report have been verified and found satisfactory.

Project Internal Examiner

Ms.V.SIRI

Assistant professor

Department of CSE

Signature of HOD

Mr. D. KRISHNA KISHORE

Associate Professor and HOD

Department of CSE

Internal Examiner

External Examiner

St. MARTIN'S ENGINEERING COLLEGE

SHAPING VIKSIT BHARAT'S BRIGHTEST MINDS

UGC Autonomous

NBA& NAAC A+ Accredited Dhulapally, Secunderabad - 500 100

www.smec.ac.in

COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I, the student of "**Bachelor of Technology in Department of Computer Science and Engineering"**, session: 2024-2028, **St. Martin's Engineering College, Dhulapally, Kompally, Secunderabad** , here by declare that the work presented in this project work is **SMART WATER CONSUMPTION MONITORING DASHBOARD** is the outcome of our own Bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of Engineering Ethics. This result embodied in this project report has not been submitted in any university for award of any degree.

V.MIDHUNA- 24K81A05R2

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose encouragement and guidance have crowded our efforts with success.

First and foremost, I would like to express my deep sense of gratitude and indebtedness to our College Management for their kind support and permission to use the facilities available in the Institute.

I especially would like to express my deep sense of gratitude and indebtedness to **Dr. K. RAVINDRA,** Director, Principal, Professor St. Martin's Engineering College Dhulapally, for permitting us to undertake this project.

I am also thankful **Mr.D. KRISHNA KISHORE,** Head of the Department of Computer Science and Engineering, St. Martin's Engineering College, Dhulapally, Secunderabad, for his support and guidance throughout my project.

I would like to express my sincere gratitude,thanks and indebtedness to my project supervisor **Ms. V. SIRI** Assistant Professor, Department of Computer Science and Engineering, St. Martins Engineering College, Dhulapally, for her support and guidance throughout project.

Finally, I express thanks to all those who have helped me in successfully completing this project. Furthermore, I would like to thank my family and friends for their moral support and encouragement. I express thanks to all those who have helped us in successfully completing the project.

V.MIDHUNA - 24K81A05R2

CONTENTS

1. ABSTRACT 1

2. INTRODUCTION 2

3. LITERATURE SURVEY 3

4. SYSTEM REQUIREMENT 4

5. FLOW CHART 5

6. ALGORITHM 6

7. SOURCE CODE 9

8. EXPERIMENTAL RESULTS 10

9. CONCLUSION 11

10.FUTURE ENHANCEMENT 12

11.REFERENCES 13

1. ABSTRACT

Water is one of the most important natural resources for human life, agriculture, industry, and environmental balance, but it is often wasted because users do not clearly understand how much they consume. The project “Smart Water Consumption Monitoring Dashboard” proposes a simple web-based application that helps users record, monitor, and understand water usage through a clear digital dashboard. The system allows users to enter the date, location or activity, and quantity of water consumed in litres. It automatically calculates total and average consumption, displays the recorded readings in a history table, presents usage trends through a chart, and shows a warning when consumption exceeds a selected limit. The prototype is developed using HTML, CSS, and JavaScript. HTML creates the structure of the dashboard, CSS provides a simple and responsive design, and JavaScript validates user input, performs calculations, updates the table and chart, and saves demonstration readings in browser storage. The application can run in a modern web browser without specialised hardware, making it suitable for an academic project demonstration. It can be useful for households, hostels, offices, and educational institutions by helping users identify high-consumption periods and make informed decisions about saving water. The current version uses manually entered readings and does not directly measure water flow or detect physical leaks. In the future, it can be enhanced with smart water meters, IoT sensors, a central database, secure user accounts, automated notifications, and detailed reports. The project demonstrates how a simple digital monitoring tool can improve awareness of water consumption and encourage responsible use of this valuable resource.

2. INTRODUCTION

Water is an essential resource for people, agriculture, industries, and the environment, but increasing demand and careless usage can lead to unnecessary wastage. In homes, hostels, colleges, and offices, water may be wasted through leaking taps, overflowing tanks, and a lack of awareness about daily consumption. Traditional monitoring methods often require users to check meter readings manually, which can make it difficult to understand usage patterns and identify sudden increases. The project "Smart Water Consumption Monitoring Dashboard" aims to provide a simple digital platform for recording and viewing water consumption in one place. Users can enter the date, location or activity, and quantity of water used in litres, while the system automatically calculates total and average consumption, displays records in a table, presents usage trends through a chart, and provides a warning when a selected limit is exceeded. The application is developed using HTML, CSS, and JavaScript, making it easy to run in a modern web browser without specialised hardware. Its main objective is to improve awareness, help users compare readings over time, and encourage responsible water use through clear information. The current prototype relies on manually entered readings and browser storage, so it does not directly measure water flow or automatically detect physical leaks. However, it provides a foundation for future improvements such as connecting smart water meters or Internet of Things sensors, storing readings in a central database, sending notifications, and generating daily or monthly reports. By presenting water-consumption information in an understandable format, the project demonstrates how a simple digital dashboard can support better monitoring and promote water conservation.

.

3. L ITERATURE SURVEY

A literature survey reviews existing ideas and technologies related to water monitoring and identifies the features that can be included in the proposed project. Traditional water monitoring usually depends on a mechanical or digital meter. These meters provide a reading of total water use, but a person may need to record readings manually and calculate differences to understand daily consumption. This method can work for basic needs, although it may be inconvenient when many readings must be compared over time.

Digital utility portals and smart meters improve access to usage information. Some systems present historical consumption, billing details, and notifications. Their benefits include easier tracking and the possibility of identifying unusual increases. However, availability depends on the utility provider, meter type, and local infrastructure. A student project cannot assume that such services are accessible to every user, so a self-contained prototype is useful for demonstrating the concept.

Internet of Things (IoT) research commonly combines flow sensors, microcontrollers, communication networks, and a software platform. A flow sensor measures water movement, while a microcontroller processes pulses or sensor signals. Readings can then be sent to a server or dashboard through Wi-Fi or another communication method. Such systems can support near-real-time monitoring, but they require hardware, calibration, stable connectivity, and attention to electrical and water safety. Sensor accuracy and installation quality influence the reliability of results.

Data visualisation is another important area. Line charts help users understand changes over time, while summary cards show key values quickly. Tables provide detailed records, and threshold alerts draw attention to readings that exceed expected limits. A well-designed dashboard should avoid clutter and label units clearly, especially litres and dates. Input validation is also necessary when readings are entered manually.

Based on these ideas, the proposed project combines simple data entry, automatic calculations, a history table, a consumption chart, and a configurable limit. It is intentionally designed as a lightweight web prototype rather than a full IoT installation. Its limitation is that manually entered readings depend on the user and browser storage is not suitable for shared, long-term production records. Future work can introduce a sensor, server database, secure accounts, and more advanced analysis. The literature review therefore supports a gradual development approach: first build a usable dashboard, then improve data collection and reliability.

4. SYSTEM REQUIREMENT

Hardware Requirements:

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB or above
- Storage: 10 GB free space
- Laptop or desktop computer for development and testing
- Android phone or emulator for mobile browser testing (optional)

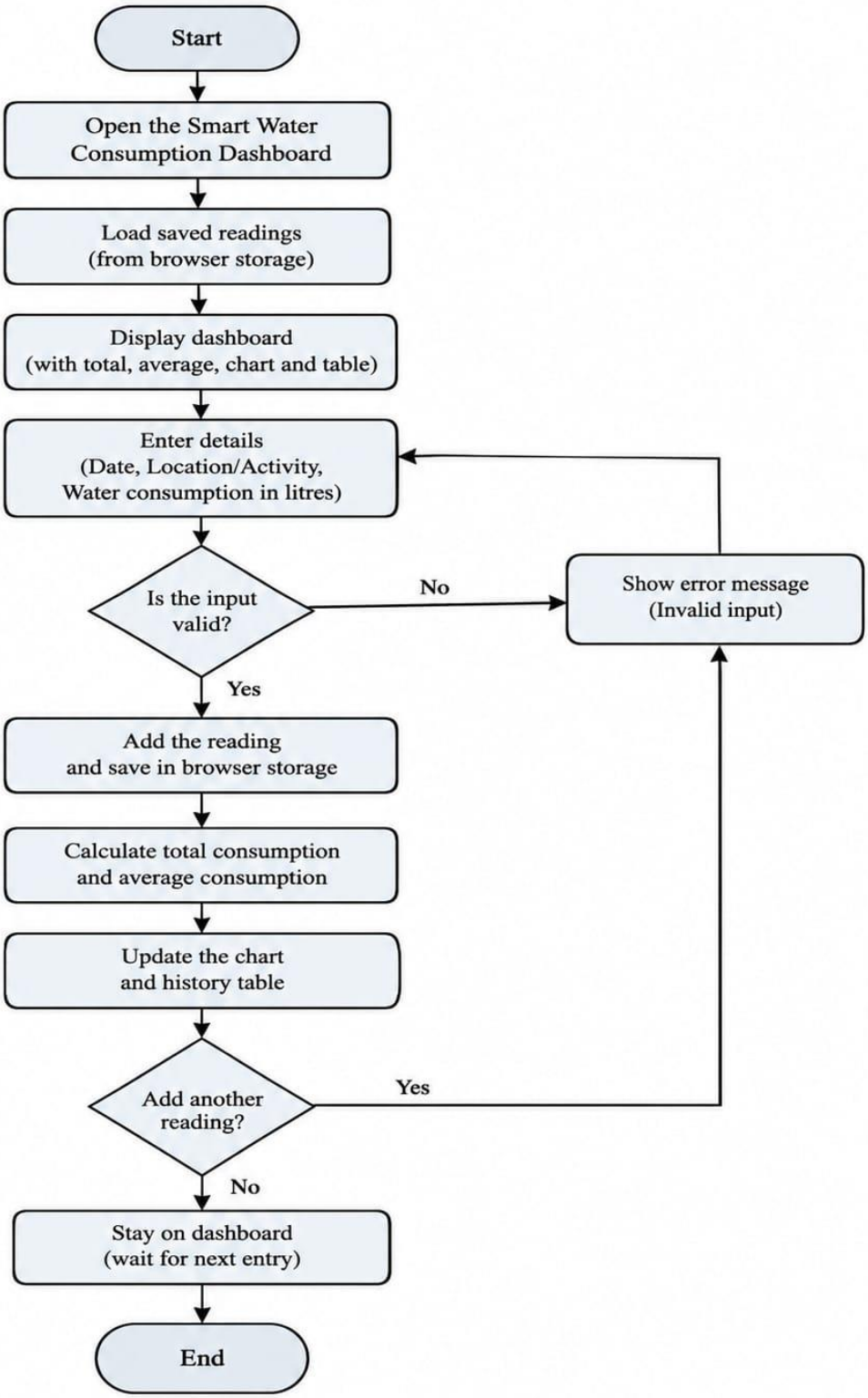
Software Requirements:

- Operating System: Windows / Linux / macOS
- HTML for webpage structure
- CSS for user interface design
- JavaScript for calculations and dashboard functionality
- Visual Studio Code or any suitable code editor
- Google Chrome, Mozilla Firefox, or another modern web browser

Functional Requirements:

- Enter water consumption readings manually.
- Enter the date and location or activity.
- Validate water consumption values.
- Calculate total and average water consumption.
- Display water usage history in a table.
- Display water consumption trends using a chart.
- Set a water consumption limit and display warnings when it is exceeded.
- Add and delete water consumption records.
- Save readings in browser storage.
- Provide a simple and user-friendly interface.

5.FLOW CHART


Start
Open the Smart Water Consumption Dashboard
Load saved readings (from browser storage)
Display dashboard (with total, average, chart and table)
Enter details (Date, Location/Activity, Water consumption in litres)
Is the input valid?
No
Show error message (Invalid input)
Yes
Add the reading and save in browser storage
Calculate total consumption and average consumption
Update the chart and history table
Add another reading?
Yes
No
Stay on dashboard (wait for next entry)
End

6. ALGORITHM

The algorithm describes the logical steps followed by the dashboard to process water-consumption readings. It is independent of a specific programming language and can be implemented in JavaScript or another suitable language. The main input consists of a date, a description of the usage location or activity, and the amount of water consumed in litres. The output includes updated totals, averages, a chart, a history table, and a status message.

1. Start the application.
2. Create an empty list for readings.
3. Read any previously saved readings from browser storage.
4. Display the dashboard with the current list.
5. Accept the date, description, and consumption quantity from the user.
6. Check that the date and description are not empty.
7. Convert the quantity to a number.
8. If the quantity is missing, invalid, or less than or equal to zero, display an error and return to Step 5
9. Create a record containing the date, description, and quantity.
10. Add the record to the readings list.
11. Save the updated readings list in browser storage.
12. Set total consumption to zero.
13. Add the quantity of every record to the total.
14. If the number of records is greater than zero, calculate average consumption as total divided by the number of records. Otherwise, set the average to zero.
15. Display the total and average with the unit “litres”.
16. Show each record in the history table.
17. Group or arrange records by date and update the chart to show the recorded quantities.
18. Compare the usage amount with the selected threshold.
19. If the threshold is exceeded, display a warning; otherwise, display a normal status.
20. If the user deletes a record, remove it from the list, save the updated list, and repeat Steps 12–19.
21. Wait for another entry or finish when the page is closed.

7.SOURCE CODE

```
import 'package:flutter/material.dart';

void main() {
  runApp(const WaterMonitoringApp());
}class WaterMonitoringApp extends StatelessWidget {
  const WaterMonitoringApp({super.key});
 @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Smart Water Monitoring',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: Colors.teal,
        ),
        useMaterial3: true,
      ),
      home: const WaterDashboard(),
    );
  }
}
class WaterReading {
  final DateTime date;
  final String location;
  final double litres;

  WaterReading({
    required this.date,
    required this.location,
    required this.litres,
  });
}class WaterDashboard extends StatefulWidget {
  const WaterDashboard({super.key});
@override
  State<WaterDashboard> createState() => _WaterDashboardState();
}class _WaterDashboardState extends State<WaterDashboard> {
  final List<WaterReading> readings = [];

  double dailyLimit = 500;

  double get totalUsage =>
      readings.fold(0.0, (sum, item) => sum + item.litres);

  double get averageUsage =>
      readings.isEmpty ? 0 : totalUsage / readings.length;

  double get todayUsage {
    final now = DateTime.now();
```

```
  return readings
      .where((item) =>
        item.date.year == now.year &&
        item.date.month == now.month &&
        item.date.day == now.day)
      .fold(0.0, (sum, item) => sum + item.litres);
  }String formatDate(DateTime date) {
   return '${date.day.toString().padLeft(2, '0')}/'
      '${date.month.toString().padLeft(2, '0')}/'
      '${date.year}';
  }Future<void> addReading() async {
   final locationController = TextEditingController();
   final litresController = TextEditingController();
   DateTime selectedDate = DateTime.now();
final result = await showDialog<WaterReading>(
     context: context,
     builder: (dialogContext) {
      return StatefulBuilder(
       builder: (context, setDialogState) {
        return AlertDialog(
         title: const Text('Add Water Reading'),
         content: SingleChildScrollView(
          child: Column(
           mainAxisSize: MainAxisSize.min,
           children: [
            ListTile(
             contentPadding: EdgeInsets.zero,
             leading: const Icon(Icons.calendar_month),
             title: Text(formatDate(selectedDate)),
             subtitle: const Text('Select date'),
             onTap: () async {
              final picked = await showDatePicker(
               context: context,
               initialDate: selectedDate,
               firstDate: DateTime(2020),
               lastDate: DateTime(2100),
              );if (picked != null) {
               setDialogState(() {
                selectedDate = picked;
               });
              }
             },
            ),
            const SizedBox(height: 12),
            TextField(
             controller: locationController,
             decoration: const InputDecoration(
              labelText: 'Location or activity',
              hintText: 'e.g. Kitchen',
              border: OutlineInputBorder(),
             ),),const SizedBox(height: 12),
```

```
TextField(
                controller: litresController,
                keyboardType:
                   const TextInputType.numberWithOptions(
                  decimal: true,
                ),decoration: const InputDecoration(
                  labelText: 'Water consumed (litres)',
                  border: OutlineInputBorder(),
                ), ),], ), ),
          actions: [
            TextButton(
              onPressed: () =>
                 Navigator.pop(dialogContext),
              child: const Text('Cancel'),
            ),
            FilledButton(
              onPressed: () {
                final location =
                   locationController.text.trim();
                final litres =
                   double.tryParse(litresController.text); if (location.isEmpty ||
                   litres == null ||
                   !litres.isFinite ||
                   litres <= 0) {
                 ScaffoldMessenger.of(context).showSnackBar(
                   const SnackBar(
                     content: Text(
                 return;
                }Navigator.pop(
                 dialogContext,
                 WaterReading(
                   date: selectedDate,
                   location: location;litres: litres,
  if (result != null && mounted) {
    setState(() {
      readings.add(result);
    });ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(
        content: Text('Water reading added successfully.'),
      ));void deleteReading(int index) {
  setState(() {
    readings.removeAt(index);
  });
 }Widget summaryCard(String title,
  String value,
  IconData icon,
 return Card(elevation: 2,const SizedBox(height: 80),
        ],
      ),
    ),
  );}}
```

8. EXPERIMENTAL RESULTS



FIG:2

9. CONCLUSION

The Smart Water Consumption Monitoring Dashboard provides a simple and effective way to monitor and manage water usage through a Flutter-based application. The system allows users to manually enter water consumption readings, view total and average usage, maintain a history of records, and visualize consumption trends through a dashboard. It also includes a daily water usage limit and displays an alert when consumption exceeds the set limit. The application helps users understand their water usage patterns, encourages responsible water consumption, and promotes water conservation. This project demonstrates how mobile application development can support efficient resource management through a user-friendly interface. In the future, the system can be enhanced with real-time water sensors, cloud storage, user authentication, and notifications to provide more accurate and advanced water monitoring.

.

10. FUTURE ENHANCEMENTS

The Smart Water Consumption Monitoring Dashboard can be enhanced in several ways to improve its functionality, accuracy, and usability. In the future, the application can be integrated with IoT-based water flow sensors to collect real-time water consumption data automatically instead of requiring users to enter readings manually. Cloud database integration can be implemented to store water usage records securely and allow users to access their data from multiple devices. User authentication can also be added to provide personalized dashboards for different households, apartments, or organizations. The system can include daily, weekly, monthly, and yearly reports to help users analyze their water consumption patterns over time. Smart notifications and alerts can be introduced to inform users when water usage exceeds a predefined limit or when unusual consumption indicates a possible leakage. Artificial intelligence and machine learning techniques could be used to predict future water requirements and identify abnormal usage patterns. Additionally, the application can provide water-saving suggestions based on individual consumption habits and compare usage with recommended limits. Support for multiple languages, improved accessibility, and a more interactive dashboard can make the application easier for different users to operate. These enhancements can transform the current prototype into a more advanced water management solution that encourages conservation, reduces wastage, and supports the sustainable use of water resources.

11. REFERENCES

1. Flutter Documentation, Google. Flutter: Build Apps for Any Screen. Available at: https://docs.flutter.dev/

2. Dart Documentation, Google. Dart Programming Language Documentation. Available at: https://dart.dev/

3. Flutter API Reference. Flutter Widgets and Material Design Components. Available at: https://api.flutter.dev/

4. World Health Organization (WHO). Domestic Water Quantity, Service Level and Health, Second Edition, 2020. Available at: https://www.who.int/publications/i/item/9789240015241

5. Flutter Widget Catalog. Official Reference for Flutter UI Components. Available at: https://docs.flutter.dev/reference/widgets

6. Visual Studio Code Documentation. Code Editing and Development Environment. Available at: https://code.visualstudio.com/docs

7. Google. Android Developers Documentation. Available at: https://developer.android.com/

8. Project implementation and testing of the Smart Water Consumption Monitoring Dashboard using Flutter and Dart.